1
INTRODUCTION TO NUTRITION

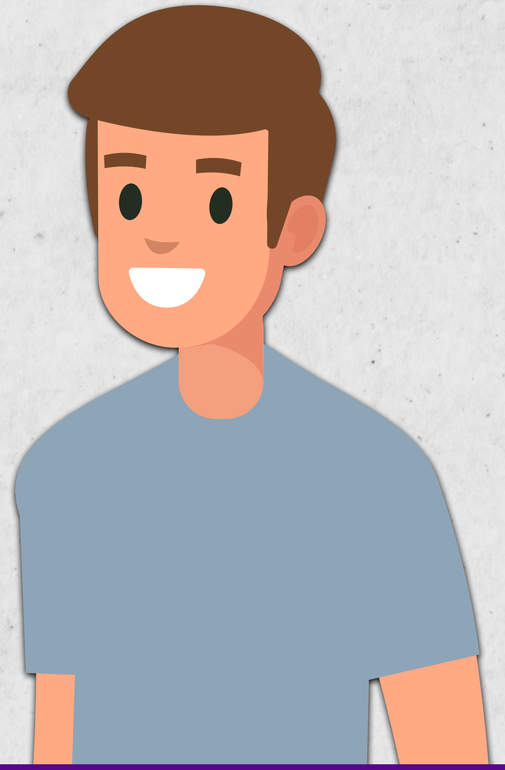
LEARNING OBJECTIVES

01 Define nutrition and explain why it is important to study it

02 List the six classes of nutrients, and identify major roles of each class of nutrient in the body

03 Discuss the factors that influence a person's food, beverage, and lifestyle choices

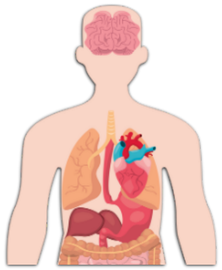
04 Examine six Key Nutrition Concepts



WHAT IS NUTRITION?

- **Nutrition is the scientific study of nutrients and how the body uses them**
  - ➡ Nutrients are the life-sustaining substances in food

Food

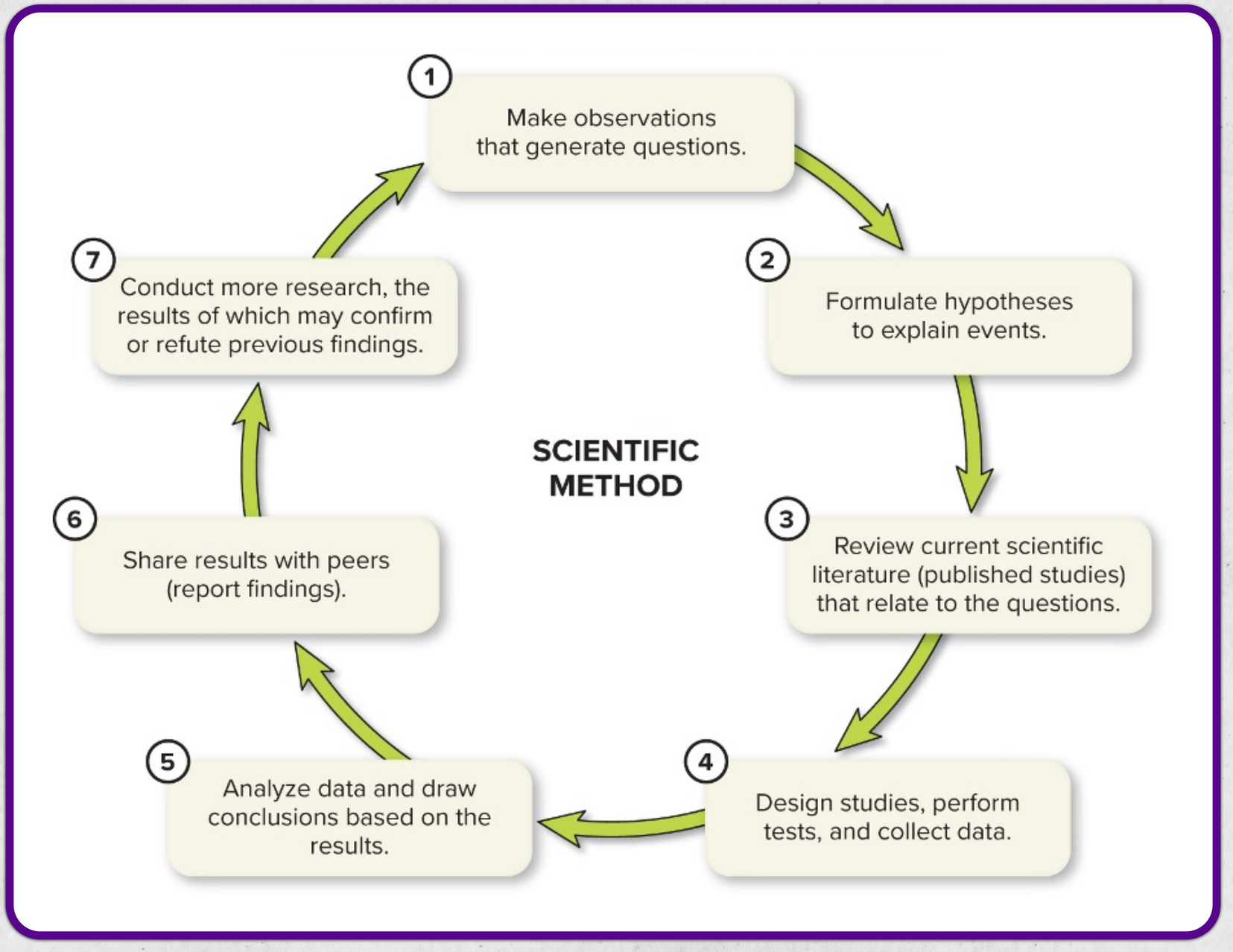


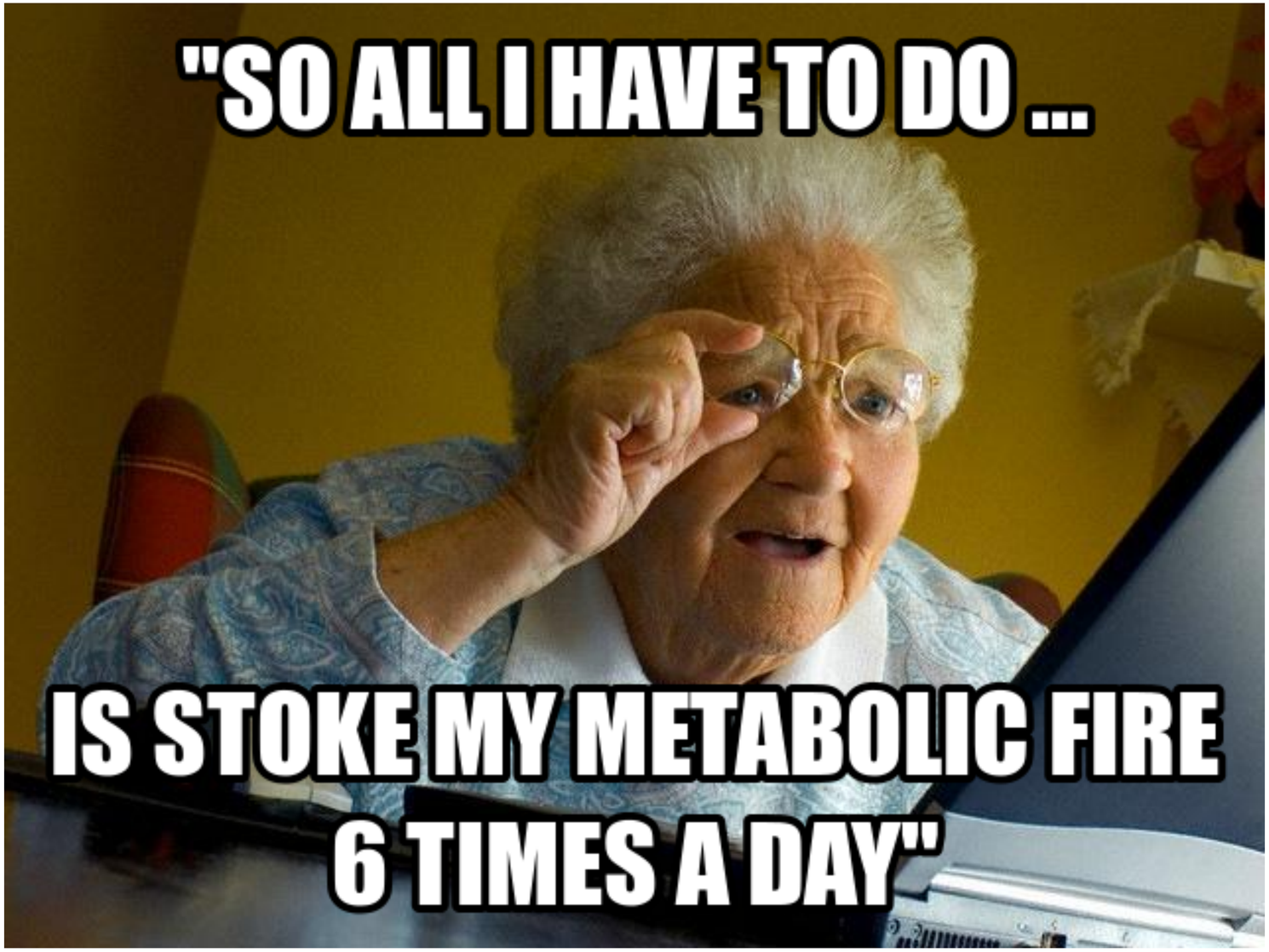
The Body

Human Behavior

Diet

"A person's usual pattern of food choices"


SCIENTIFIC METHOD
1
Make observations that generate questions.
2
Formulate hypotheses to explain events.
3
Review current scientific literature (published studies) that relate to the questions.
4
Design studies, perform tests, and collect data.
5
Analyze data and draw conclusions based on the results.
6
Share results with peers (report findings).
7
Conduct more research, the results of which may confirm or refute previous findings.

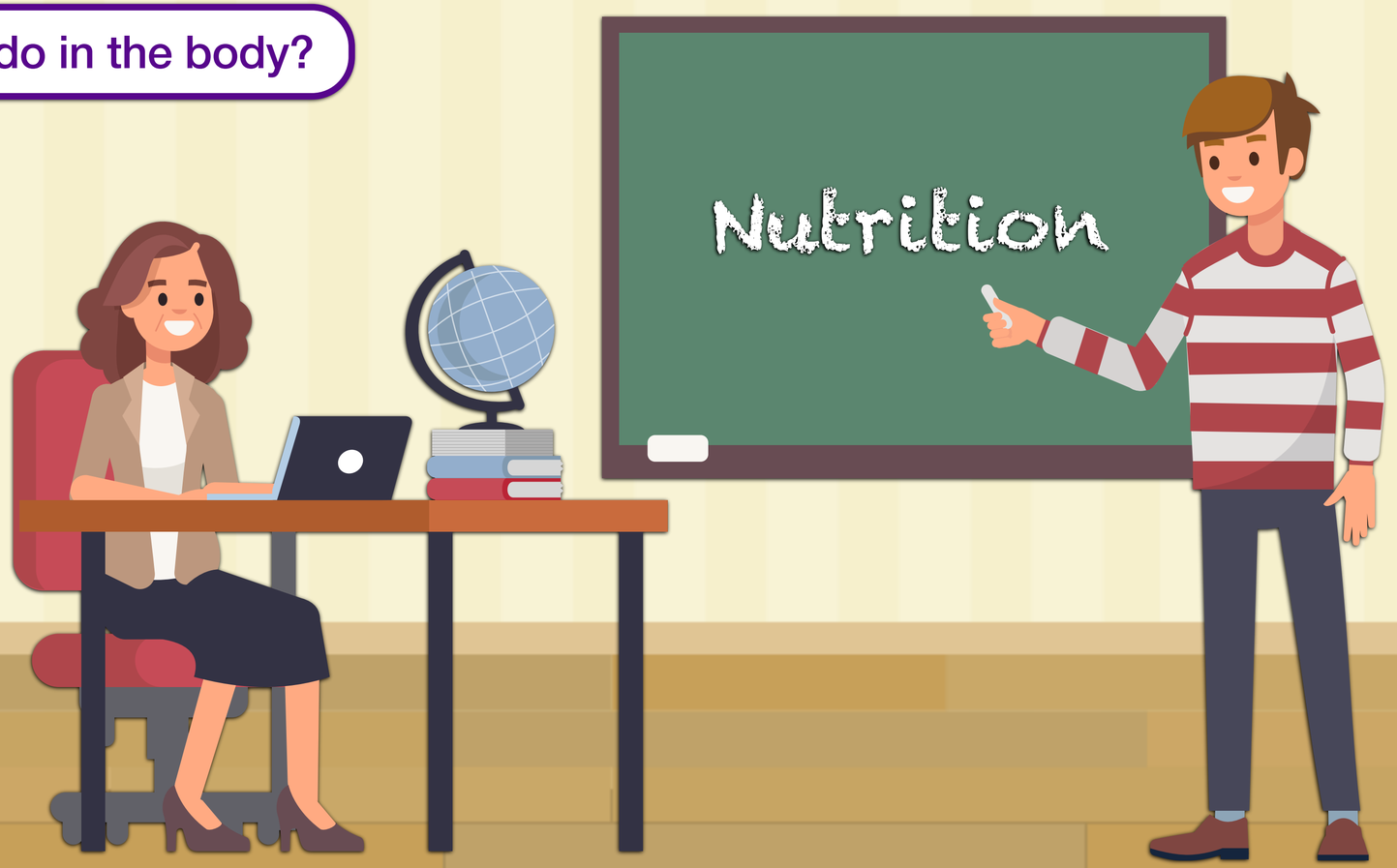

"SO ALL I HAVE TO DO ...
IS STOKE MY METABOLIC FIRE
6 TIMES A DAY"

Which nutrients does a food contain?

How much of each nutrient do humans need to consume?

What happens if someone consumes too much/little of a specific nutrient?

What do nutrients do in the body?



WHY STUDY NUTRITION?

- **Food is a basic human need for survival**
  - ➡ Nutrients are necessary for the growth, maintenance, and repair of tissues
- **Failure to make appropriate food selections can be harmful to health**
  - ➡ Diet is influential in a person's risk of developing many chronic diseases
- **Several of the 10 leading causes of deaths in the U.S. are related to diet**
  - ➡ Examples: heart disease and diabetes

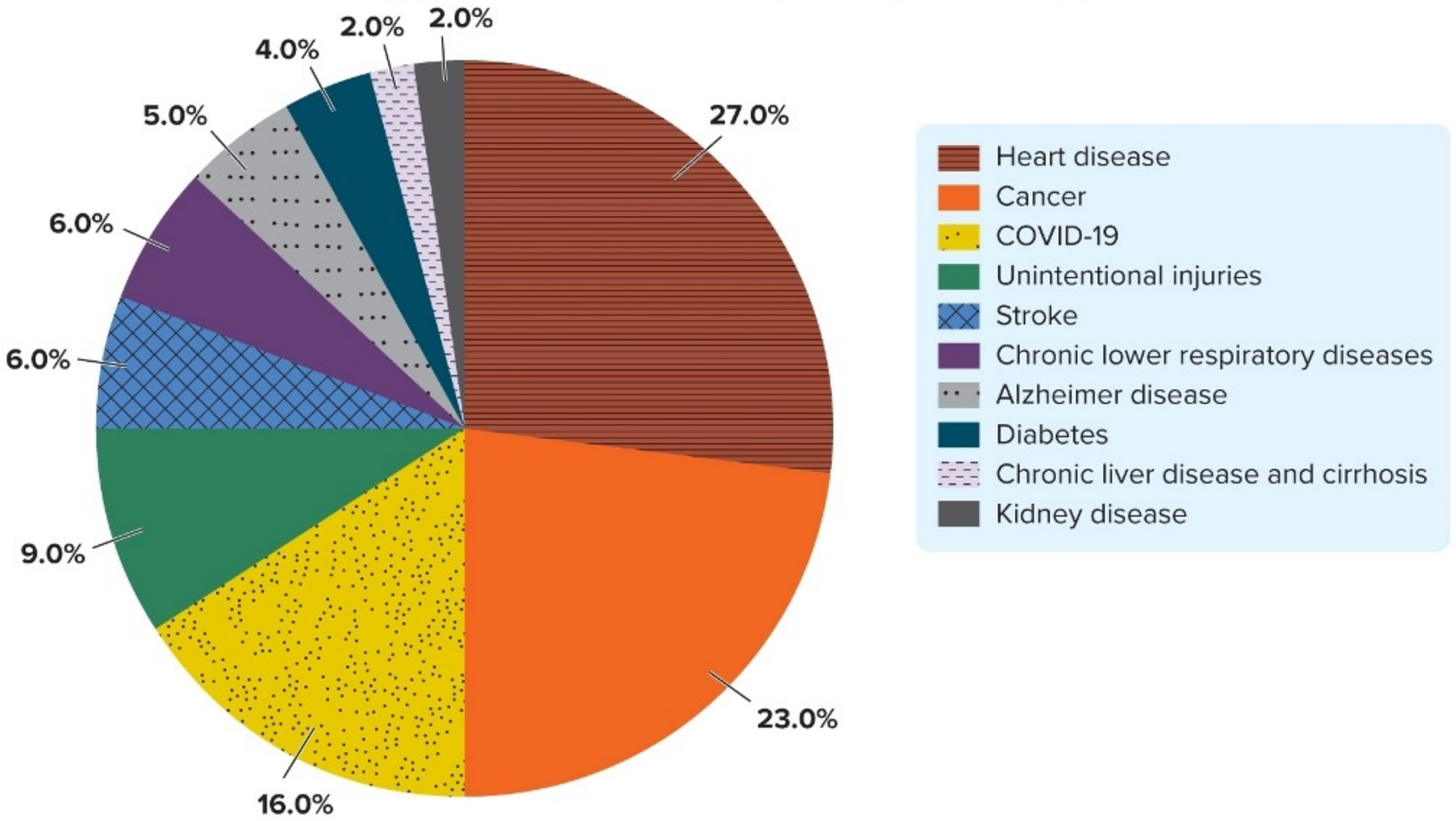

27.0%
23.0%
16.0%
9.0%
6.0%
6.0%
5.0%
4.0%
2.0%
2.0%
Heart disease
Cancer
COVID-19
Unintentional injuries
Stroke
Chronic lower respiratory diseases
Alzheimer disease
Diabetes
Chronic liver disease and cirrhosis
Kidney disease

Figure 1. Life's Essential 8. Life's Essential 8 are the key measures for improving and maintaining cardiovascular health as defined by the American Heart Association.

A NOTE ON GENETICS AND LIFESTYLE

- **Most chronic diseases cannot be prevented by diet alone**
  - ➡ **Other influential risk factors include:**
    - * *Genetics*
    - * *Environment*
    - * *Sleep*
    - * *Drug and Tobacco Use*
    - * *Physical Activity*
    - * *Stress*

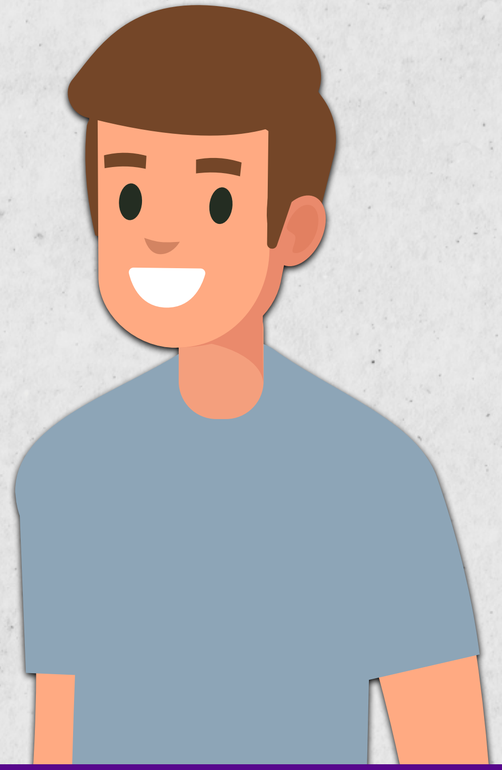
LEARNING OBJECTIVES

01 Define nutrition and explain why it is important to study it

02 List the six classes of nutrients, and identify major roles of each class of nutrient in the body

03 Discuss the factors that influence a person's food, beverage, and lifestyle choices

04 Examine six Key Nutrition Concepts



NUTRIENTS

- **Components of food that are required for the body to function properly**
  - ➡ Carbohydrates
  - ➡ Fat
  - ➡ Protein

  → **Macronutrients**

  - ➡ Vitamins
  - ➡ Minerals

  → **Micronutrients**

  - ➡ Water
- **Most nutrients have more than one physiological role or function in the body**
  - ➡ Energy, growth, and development
  - ➡ Regulation of processes (i.e. repair and maintenance of cells)

In a way, you really are what you eat…

Male
Female
Carbohydrate
<1%
<1%
Minerals
6%
5%
Protein
16%
13%
Fat
16%
25%
Water
62%
57%

Nutrient Class	Major Functions
Carbohydrates	• Major source of energy (most forms) • Maintenance of normal blood glucose levels • Elimination of solid waste from gastrointestinal tract (fiber)
Lipids	• Major source of energy • Cellular development, physical growth and development • Regulation of body processes (certain hormones, for example) • Growth and development of the brain • Absorption of fat-soluble vitamins
Proteins	• Formation of structural components, such as muscle fibers, and functional components, such as enzymes • Cellular development, growth, and maintenance • Regulation of body processes (certain hormones, for example) • Transportation of substances within the blood • Energy (normally a minor source)
Vitamins	• Regulation of body processes • Immune function • Production and maintenance of cells • Protection against agents that can damage cellular components

Nutrient Class	Major Functions
Minerals	• Regulation of body processes, including fluid balance and energy metabolism • Formation of certain chemical messengers • Formation of structural and functional components of various substances and tissues • Cellular development, growth, and maintenance
Water	• Maintenance of fluid balance • Regulation of body temperature • Elimination of wastes • Transportation of substances • Participation in many chemical reactions

99.9%

Oxygen
Carbon
Hydrogen
Nitrogen
Calcium
Phosphorus
Potassium
Sulfur
Chlorine
Sodium
Magnesium

Less than 0.1%

Cobalt
Copper
Fluorine
Iodine
Iron
Manganese
Zinc
Selenium
Molybdenum
Chromium

Nutrient Class	Carbon	Hydrogen	Nitrogen	Oxygen	Phosphorus	Sulfur	Others
Carbohydrates	✓	✓	-	✓	-	-	-
Lipids	✓	✓	X	✓	X	-	-
Proteins	✓	✓	✓	✓	X	-	-
Vitamins	✓	✓	X	✓	X	X	Cobalt
Minerals	-	-	-	X	X	-	Sodium, magnesium, potassium, calcium, chromium, iron, copper, zinc, iodine, and others
Water	-	✓	-	✓	-	-	-

ESSENTIAL NUTRIENTS

- **A nutrient that must be supplied by food**
  - ➡ If the nutrient is missing from the diet, a deficiency disease occurs
- **Non-essential nutrients are those the body produces enough of on its own**
  - ➡ Can become conditionally essential under certain circumstances

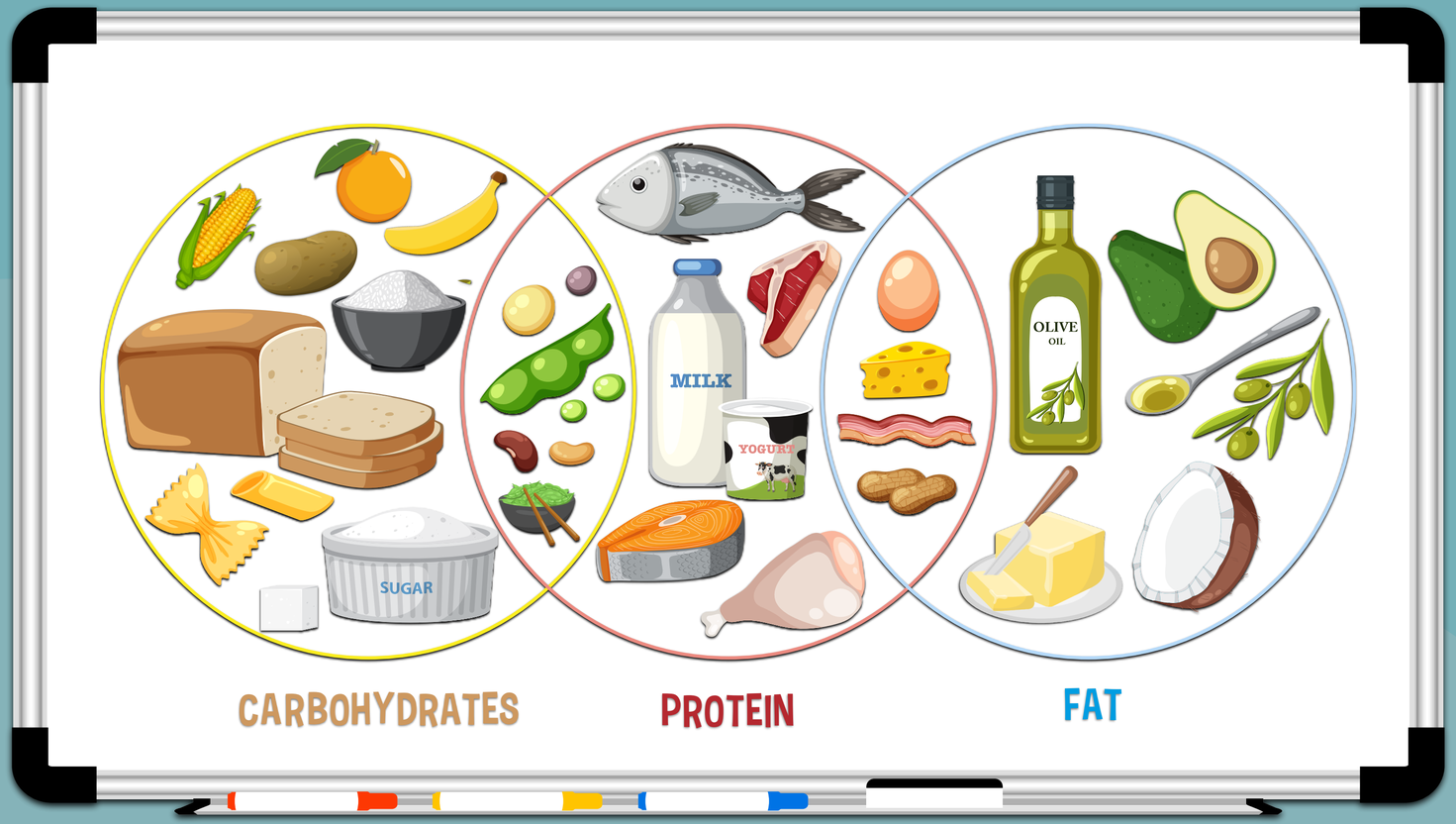
PHYTOCHEMICALS

- **Substances in plants that are not nutrients but may have health benefits**
  - ➡ Examples: beta-carotene, lycopene, and caffeine
- **Not all phytochemical have beneficial effects on the body**
  - ➡ Example: nicotine and ricin

MACRONUTRIENTS

- **Energy-yielding nutrients (or calorie-containing nutrients)**
  - ➡ **The energy the body extracts from them supports bodily functions**
    - * *Blood Circulation*
    - * *Digestion*
    - * *Breathing*
    - * *Physical Activity*
- **Measured in calories and grams**
  - ➡ **Carbohydrates = 4 calories/gram**
  - ➡ **Protein = 4 calories/gram**
  - ➡ **Fat = 9 calories/gram**

Note: Alcohol provides 7 calories/gram but it's not considered a nutrient because it interferes with the growth and repair of tissues



We often identify foods by the predominant macronutrient, but most foods contain a mix of macronutrients!

MICRONUTRIENTS

- **Support processes in the body but do <u>not</u> provide energy**
  - ➡ **Needed in smaller amounts than the macronutrients**
    - * *Milligrams*
    - * *Micrograms*
- **Includes the <u>vitamins</u> and <u>minerals</u>**
  - ➡ **Examples**
    - * *Vitamin C*
    - * *Vitamin D*
    - * *Niacin*
    - * *Folate*
    - * *Calcium*
    - * *Sodium*
    - * *Potassium*
    - * *Iron*

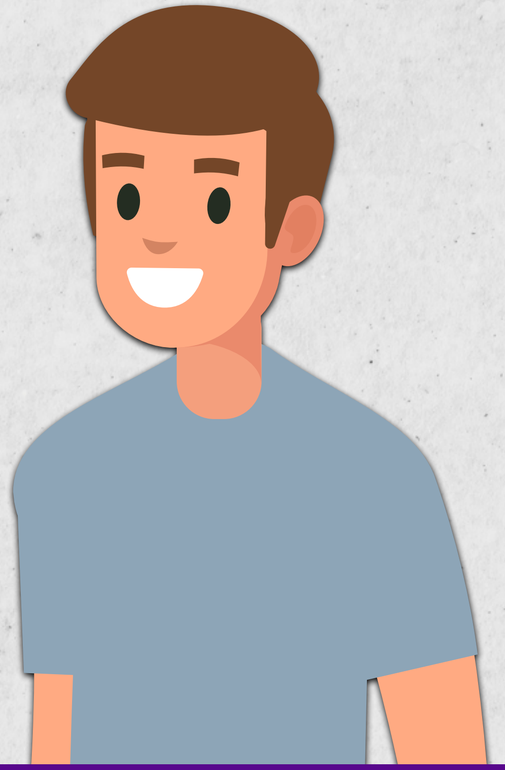
Vitamins	Minerals
Organic Substances (contain carbon)	*Inorganic Substances (don't contain carbon)*
Made by plants and animals	*Found in the Earth's crust*

MACRONUTRIENTS

MICRONUTRIENTS

LEARNING OBJECTIVES

- [x] 01 Define nutrition and explain why it is important to study it
- [x] 02 List the six classes of nutrients, and identify major roles of each class of nutrient in the body
- [ ] 03 Discuss the factors that influence a person's food, beverage, and lifestyle choices
- [ ] 04 Examine six Key Nutrition Concepts



WHAT FACTORS INFLUENCE A PERSON'S EATING PRACTICES?

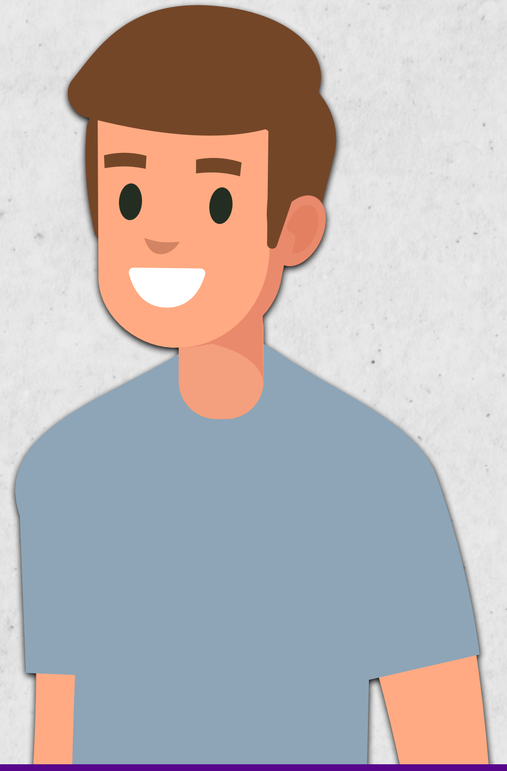
FACTORS THAT INFLUENCE EATING

1. **Life stage/age**
2. **Sensory information** (smell, taste, texture, and appearance of food)
3. **Cultural/religious practices**
4. **Emotions** (happy, sad, stressed, and bored)
5. **Past experiences**
6. **Food cost/availability**
7. **Food marketing/advertising**
8. **Medical conditions**
9. **Health education**
10. **Health goals** (weight loss, weight gain, muscle building, and performance)

Nutrition
Childhood Experiences
Beliefs
Media
Religion
Occupation
Convenience
Cost
Flavor
Habits
Education
Health
Moods
Taste

LEARNING OBJECTIVES

- 01 Define nutrition and explain why it is important to study it
- 02 List the six classes of nutrients, and identify major roles of each class of nutrient in the body
- 03 Discuss the factors that influence a person's food, beverage, and lifestyle choices
- 04 Examine six Key Nutrition Concepts



Concept	Description
Concept 1	Most foods are mixtures of nutrients.
Concept 2	Variety, moderation, and balance can help ensure a diet's nutritional adequacy.
Concept 3	Food is the best source of nutrients.
Concept 4	Foods and the nutrients they contain are one of many components of health.
Concept 5	Malnutrition includes overnutrition as well as undernutrition.
Concept 6	Nutrition is a dynamic science.

LEARNING OBJECTIVES

- 01 Define nutrition and explain why it is important to study it
- 02 List the six classes of nutrients, and identify major roles of each class of nutrient in the body
- 03 Discuss the factors that influence a person's food, beverage, and lifestyle choices
- 04 Examine six Key Nutrition Concepts